

INDUSTRIAL MANAGEMENT PART 2

- **ORGANISATION.**
- **SOCIAL UNIT OF PEOPLE SYSTEMATICALLY ARRANGED AND MANAGED TO PURSUE COLLECTIVE GOALS ON CONTUNUING BASIS.**

STRUCTRE

- **MECHANISTIC**
- **ORGANIC**

The structure defines position and function. The optimum structure depends on Political, Demogrphic, Sociocultural , Economic, Technological, AND Legal factors.

Mechanistic	Organic
Individual specialization: Employees work separately and specialize in one task	Joint Specialization: Employees work together and coordinate tasks
Simple integrating mechanisms: Hierarchy of authority well-defined	Complex integrating mechanisms: task forces and teams are primary integrating mechanisms
Centralization: Decision-making kept as high as possible. Most communication is vertical.	Decentralization: Authority to control tasks is delegated. Most communication lateral
Standardization: Extensive use made of rules & Standard Operating Procedures	Mutual Adjustment: Face-to-face contact for coordination. Work process tends to be unpredictable
Much written communication	Much verbal communication
Informal status in org based on size of empire	Informal status based on perceived brilliance
Organization is a network of positions, corresponding to tasks. Typically each person corresponds to one task	Organization is network of persons or teams. People work in different capacities simultaneously and over time

MECHANISTIC AND ORGANIC

- RIGID, WELL DEFINED, BEUROCRATIC, MULTI LEVEL, PYRAMID, LEAST INTERACTIVE AMONG PEERS.
- FLEXIBLE, FLAT, INTERACTIVE, MINIMUM DOCUMENTATION, MULTIPLE ROLES, DECENTRALIZED DECISION MAKING.

GREINERS MODEL

- **ASSUMPTIONS**

- a) GROWTH
- b) MANAGEMENT STYLE
- c) STRUCTURE
- d) COORDINATION

TYPES OF SFSTRUCTURES

- FUNCTIONAL Manufacturing firms
- DIVISIONAL Multiple markets
- MATRIX Project Based Companies
- TEAM BASED Flexibility Innovation
- NETWORK Outsourcing Companies
- HIERARCHIAL Traditional Stable
- FLAT Faster Decisions
Collaborative

PHASES OF GROWTH

GROWTH THROUGH CREATIVITY

LEADERSHIP CRISIS

GROWTH THROUGH DIRECTION

AUTONOMY CRISIS

GROWTH THROUGH DELEGATION

CONTROL CRISIS

GROWTH THROUGH COORDINATION

RED TAPE CRISIS

GROWTH THROUGH COLLABORATION

INTERNAL GROWTH CRISIS

GROWTH THROUGH ALLIANCES

IDENTITY CRISIS.

STAGE 1 CREATIVITY

Flat structure
Small and Informal
Entrepreneurial
Loyal employees.

ONE MAN SHOW...Not possible

STAGE 2 DIRECTION

Appoint Functional Managers.

**Create Middle Management, Rules procedures are standardized,
Central coordination with Entrepreneur**

Autonomy crisis is Born.

STAGE 3 DELEGATION

1. Tactical , Operational objectives by middle managers.
2. Low intervention
3. Product Group
4. Individual Managers

Co ordination Difficult Divisional managers plot own course

STAGE 4 COORDINATION

1. Staff Departments Introduced... What are staff and line roles?
2. Take strong position from Headquarters
3. Manage Divisional managers
4. Headquarters All important

COMPANY INFLEXIBLE RED TAPE ARISES STAFF DEPT TAKE UP TOO MUCH POWER

STAGE 5 COLLABORATION

1. Line and Staff Department co-ordination
2. Hierachial forms
3. Matrix Structure
4. Consultation Groups
5. Meetings

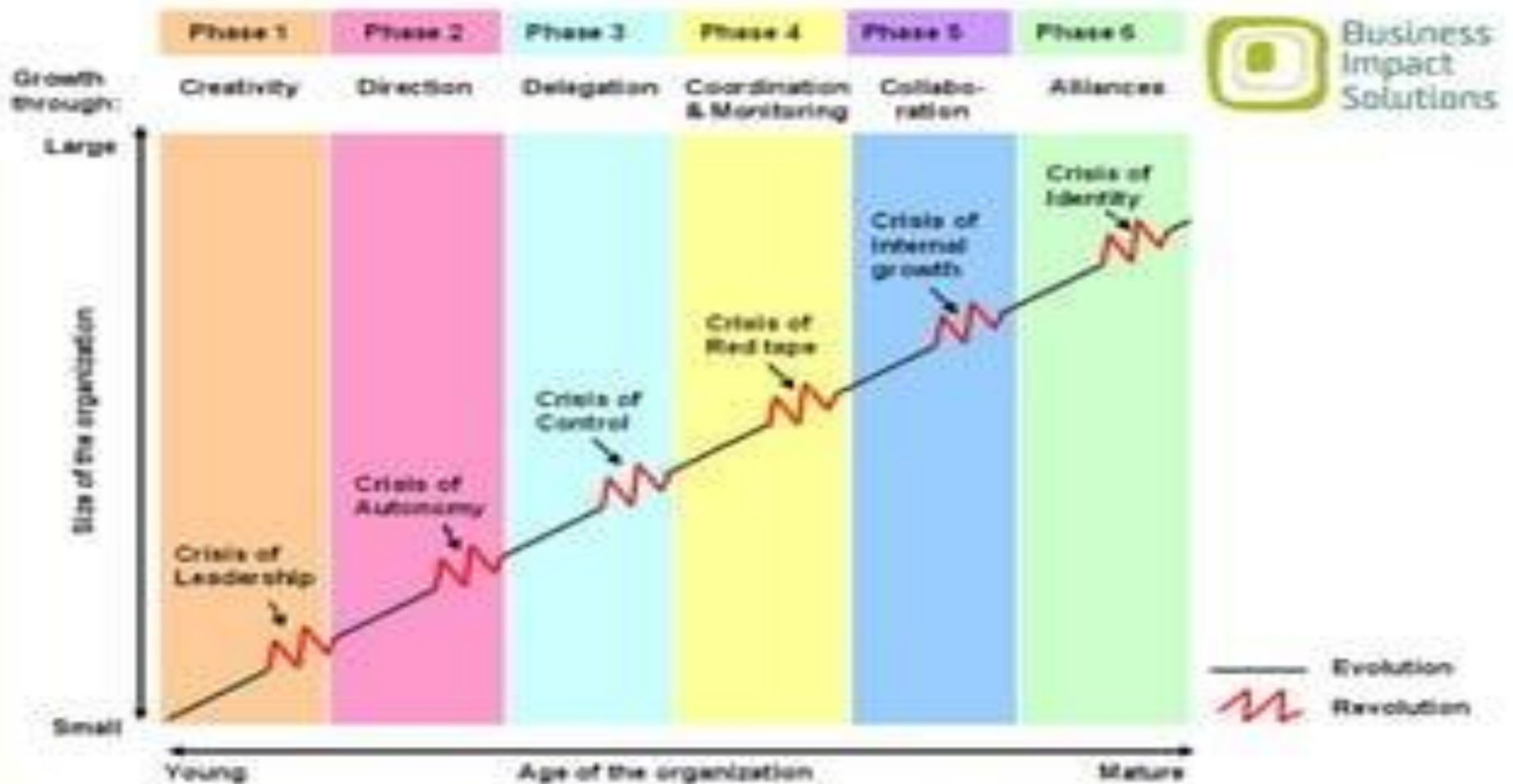
Too many Consultations. Supervision Declines...

STAGE 6 ALLIANCES

1. External alliances
2. Contacts
3. Mergers
4. Extensive Network

Focus on alliances rather than own business. Other businesses take over. Identity Crisis.

GRAPH

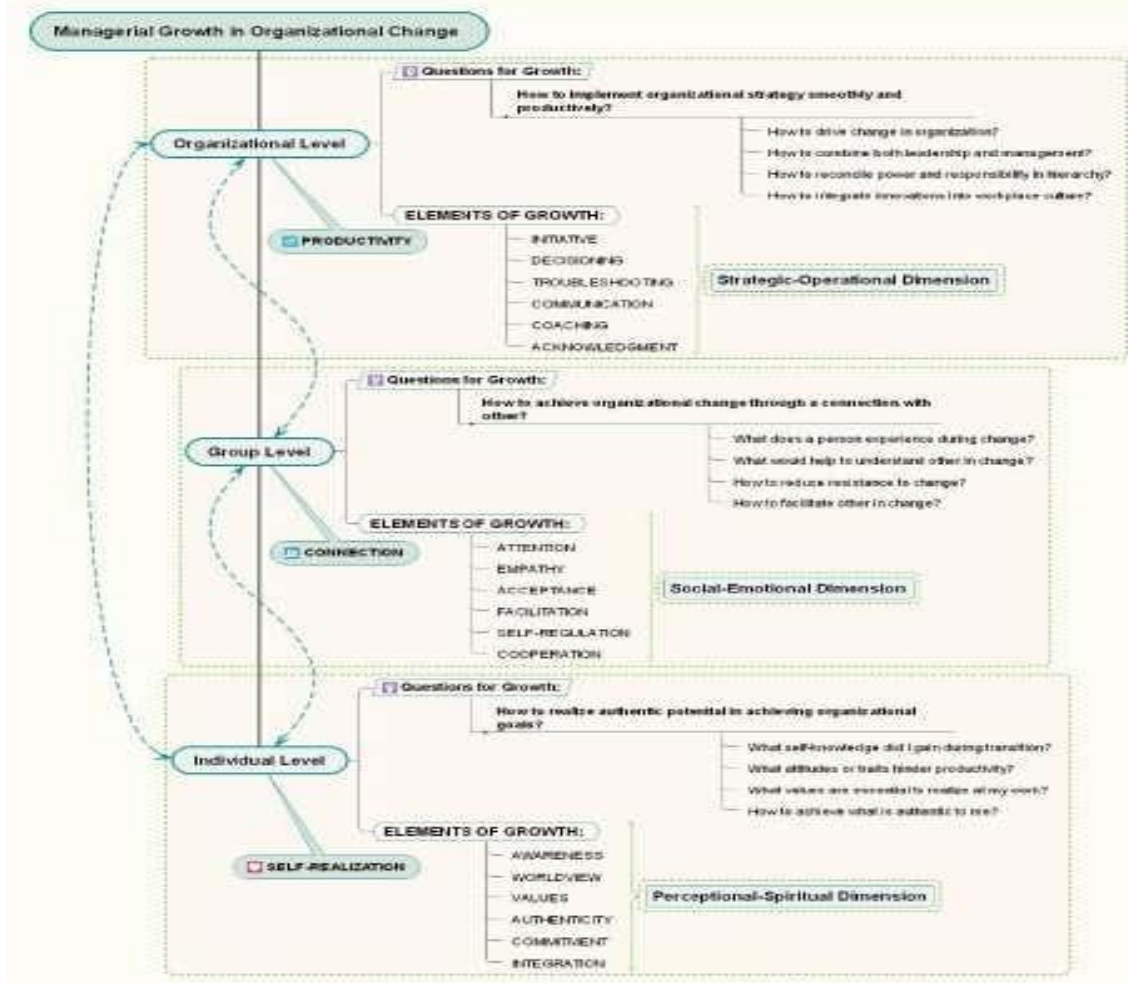


Creating & implementing insight-led marketing strategies with real commercial impact

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GRAPH 2



ORGANISATIONAL DESIGN

**CREATE VALUE FOR FUTURE
PEOPLE AT THE CENTRE
DESIGN IS LIKE A MAP. IMPLEMENTATION GPS
BUILD RESILIENCE FOR UNKNOWN.**

NO PERFECT MODEL.

ORGANISATIONAL DESIGN

MISSION

PROCESS

ROLES

STRUCTURE

PEOPLE AND PRACTICES

Concept of Internal customers.

Customer is The Focus.

**Research.> Development.> Process Design.> Projects.>
Operations.>Marketing.> Final Purchaser.**

Multiple Customers. Purchase, Finance and accounts, Personnel, Quality Assurance, HR, Maintainance, Stores, Security, IT services.

- **Interaction With Others:**

1. R&D
2. Design & projects.
3. Operations
4. Quality assurance
5. Customer

Marketing

- **Three Major Areas**

1. New Products
2. Improving Existing products
3. Service to the customer

PORTERS VALUE CHAIN MODEL

INTRODUCED IN 1985 BY MICHEL PORTER

Primary Activities- Directly Develop Product or Services.

Support Activities- To make Primary Activities more Efficient

Strategic Management tool.

Support Activities

Firm Infrastructure

Management, finance, legal, planning

Human Resource Management

Professional development, employee relations, performance appraisals, recruiting, competitive wages, training programs

Technology Development

Integrated supply chain system, real-time sales information

Procurement

Real-time inventory, Communication with suppliers, Purchase supplies and materials

Inbound Logistics

- Real-time inbound inventory data
- Location of distribution facilities
- Trucks
- Material Handling
- Warehouse

Operations

- Standardized model
- Access to real-time sales and inventory system

Outbound Logistics

- Order processing
- Full delivery trucks

Marketing and Sales

- Pricing
- Communication
- Promotion
- Products based on community needs
- Low prices

Service

- Delivery
- Installation
- Repair
- Greeters
- Customer service focus

Margin

Margin

Primary Activities

VALUE CHAIN ANALYSIS

COMPETITION

NEW ENTRANTS

BARGAINING POWER OF CUSTOMERS

BARGAINING POWER OF SUPPLIERS

THREAT OF ALTERNATIVES



PORTER'S FIVE FORCES IS A FRAMEWORK FOR UNDERSTANDING THE COMPETITIVE FORCES AT WORK IN AN INDUSTRY, AND WHICH DRIVE THE WAY ECONOMIC VALUE IS DIVIDED AMONG INDUSTRY ACTORS.

VALUE CHAIN

PHYSICAL OR TRADITIONAL

VIRTUAL

STRATEGIC BUSINESS UNIT

GLOBAL

EXAMPLES OF INDUSTRIES

WALL MART RELIANCE

AMAZON

FEDEX SPEED POST

BASF

CITI BANK ICICI BANK

ASIAN PAINTS PIDILITE

PORTERS GENERIC STRATEGY

LOWER COST

DIFFERENTIATION.

COMPETITIVE ADVANTAGE
What makes the Company "Strong"?

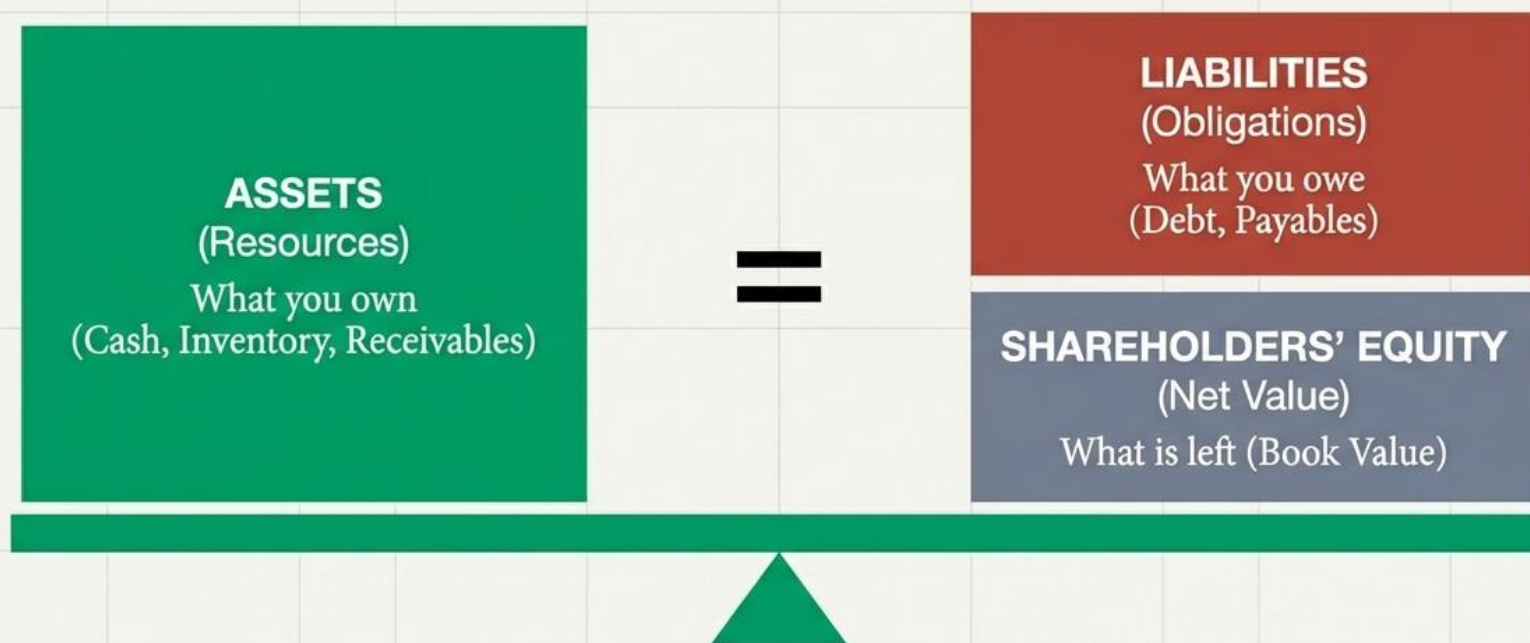
MARKET SCOPE
What part of the Market is being targeted?

	COSTS	DIFFERENTIATION
ENTIRE	COST LEADERSHIP Strategy: Be the most competitive company in Cost in the Entire Market.	DIFFERENTIATION Strategy: Be a Distinctive company, recognized for its Uniqueness, Quality or Personality.
A SEGMENT	COST FOCUS Strategy: Be very competitive in Cost in a particular Product or Niche.	DIFFERENTIATION FOCUS Strategy: Have a Differentiated Product or Market Niche.

Balance Sheet

- Reports assets, liabilities, and equity at a specific point in time
- What the company currently holds and owns
- The balance sheet is a **snapshot** that tells you what a company is worth at a single moment

The Golden Equation: Assets must always equal funding



This equation must always balance. It describes the same money from two perspectives: Where it is (Assets) and where it came from (Liabilities + Equity).

Breaking Down the Balance Sheet



ASSETS

- Current: Cash, Inventory, Accounts Receivable (Liquid)
- Non-Current: Property, Equipment, Long-term Investments



LIABILITIES

- Current: Accounts Payable, Short-term loans
- Non-Current: Long-term Debt, Deferred Tax



EQUITY

- Retained Earnings (Reinvested profits)
- Capital Stock (Initial investment)

“The amount returned to shareholders if all assets were liquidated and debts repaid.”

Balance sheet example

TEDDY FAB INC. BALANCE SHEET December 31, 2100

ASSETS

Current assets

Cash and cash equivalents	\$ 100,000
Accounts receivable	20,000
Inventory	15,000
Prepaid expense	4,000
Investments	10,000
Total current assets	149,000

Property and equipment

Land	24,300
Buildings and improvements	250,000
Equipment	50,000
Less accumulated depreciation	(5,000)

Other assets

Intangible assets	4,000
Less accumulated amortization	(200)

Total assets **\$ 472,100**

LIABILITIES AND SHAREHOLDERS' EQUITY

Current liabilities

Accounts payable	\$ 30,000
Notes payable	10,000
Accrued expenses	5,000
Deferred revenue	2,000
Total current liabilities	47,000

Long-term debt 200,000

Total liabilities **247,000**

Shareholders' Equity

Common stock	10,000
Additional paid-in capital	20,000
Retained earnings	197,100
Treasury stock	(2,000)

Total liabilities and shareholders' equity **\$ 472,100**

Case Study: Using the Balance Sheet for a solvency check

Elsie's Child Care: End of Quarter Status

Total Assets: \$14,500

- Cash in Bank + Accounts Receivable

—

Total Liabilities: \$7,500

- Credit Card Debt + Business Loan

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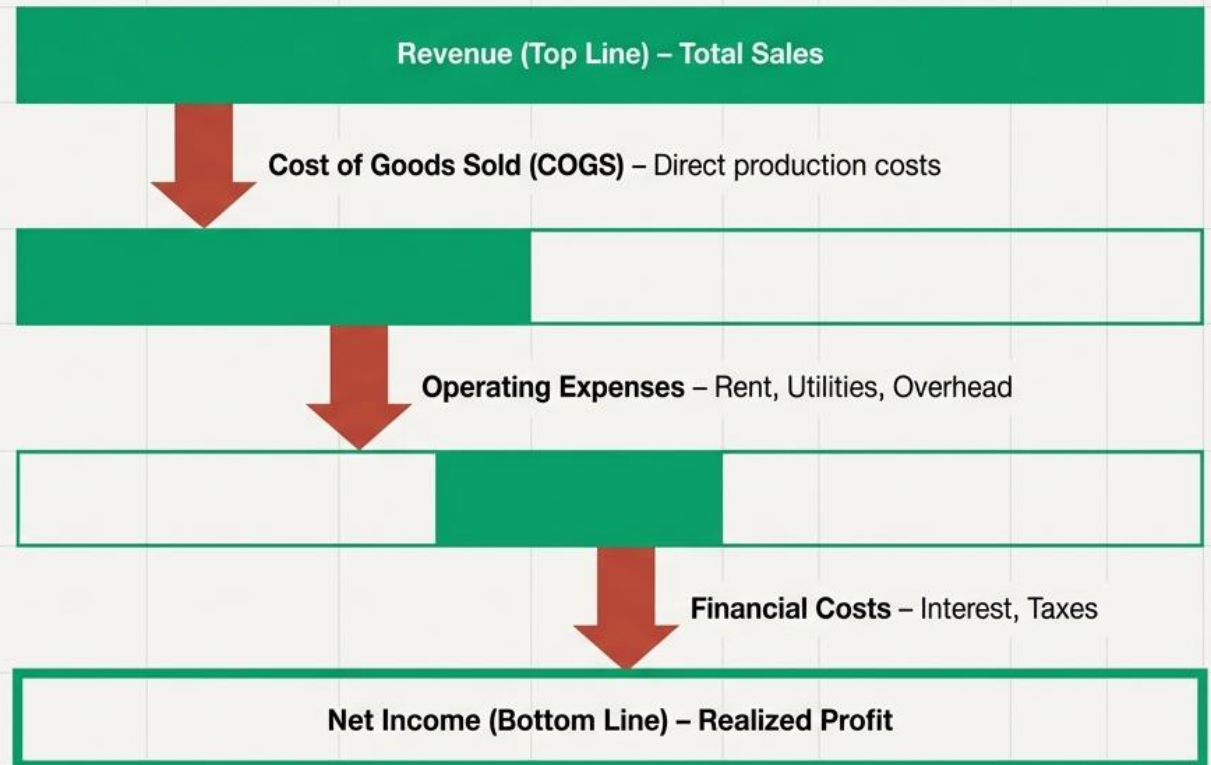
Owner's Equity: \$7,000

The Solvency Insight. Despite the 'broken pipe' hurting the P&L profit, the Balance Sheet proves Elsie is solvent. Her assets cover her debts nearly 2-to-1.

P&L Statement

- Summarises the **revenues**, **costs** and **expenses** incurred during a specific period of **time**
- Captures the flow of activity over a period of time like a **movie**, revealing the story of how it generated profit or incurred loss through its daily operations

Tracing the path from Revenue to Realized Profit



The Anatomy of a P&L.

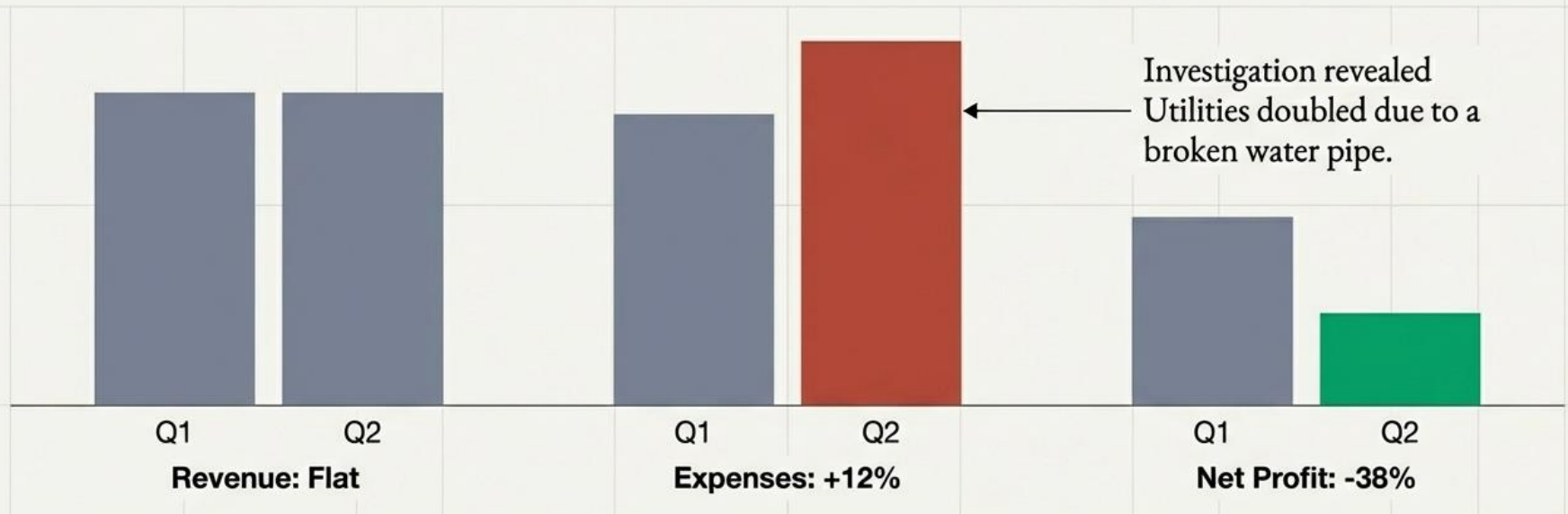
This document reveals the company's ability to turn sales into actual earnings by subtracting the costs of doing business.

Company ABC Ltd.**Profit and Loss Statement****For the year ended December 31, 2018**

Particulars	Amount (\$)	Amount (\$)
Sales	\$ 1,000,000	
Cost of goods Sold	\$ 600,000	
Gross Profit		\$ 400,000
Operating Expenses		
Advertisement expenses	\$ 60,000	
Depreciation expense	\$ 80,000	
Rent expense	\$ 15,000	
Payroll taxes	\$ 5,000	
salaries and wages	\$ 51,000	
Commission expense	\$ 5,000	
Other Operating Expenses	\$ 7,000	
Total operating Expenses		\$ 223,000
Operating Income		\$ 177,000
Non-operating Income		
Revenue from Interest	\$ 10,500	
Interest Expenses	\$ (9,100)	
Total Non- operating Income		\$ 1,400
Net Income		\$ 178,400

Case Study: How the P&L detects operational leaks

Elsie's Child Care: Quarter 1 vs. Quarter 2 Performance



The Insight: The P&L acts as a diagnostic tool. It flagged a drop in efficiency (the leak) even though top-line revenue remained stable.

Placing the documents head-to-head

	PROFIT & LOSS (P&L)	BALANCE SHEET
 TIME	Period of Time (Duration/Flow)	Specific Date (Snapshot/Stock)
 FOCUS	Profitability & Efficiency	Wealth, Liquidity & Solvency
 KEY METRIC	Net Income (Bottom Line)	Shareholder Equity

Reporting Order: Public companies list Balance Sheet first; Private companies often list P&L first.

Critical distinctions and common pitfalls



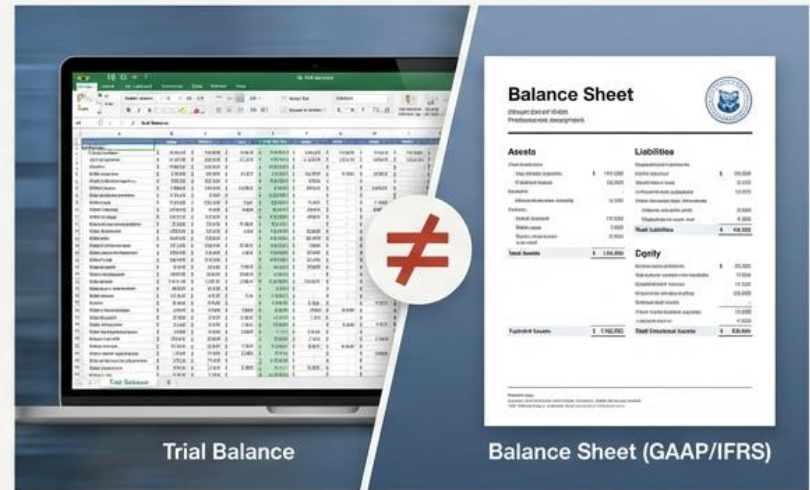
Profit $\neq$ Cash Flow

A firm can be profitable on the P&L (sales on credit) without having cash in the bank. Do not confuse “Net Income” with “Cash Balance”.

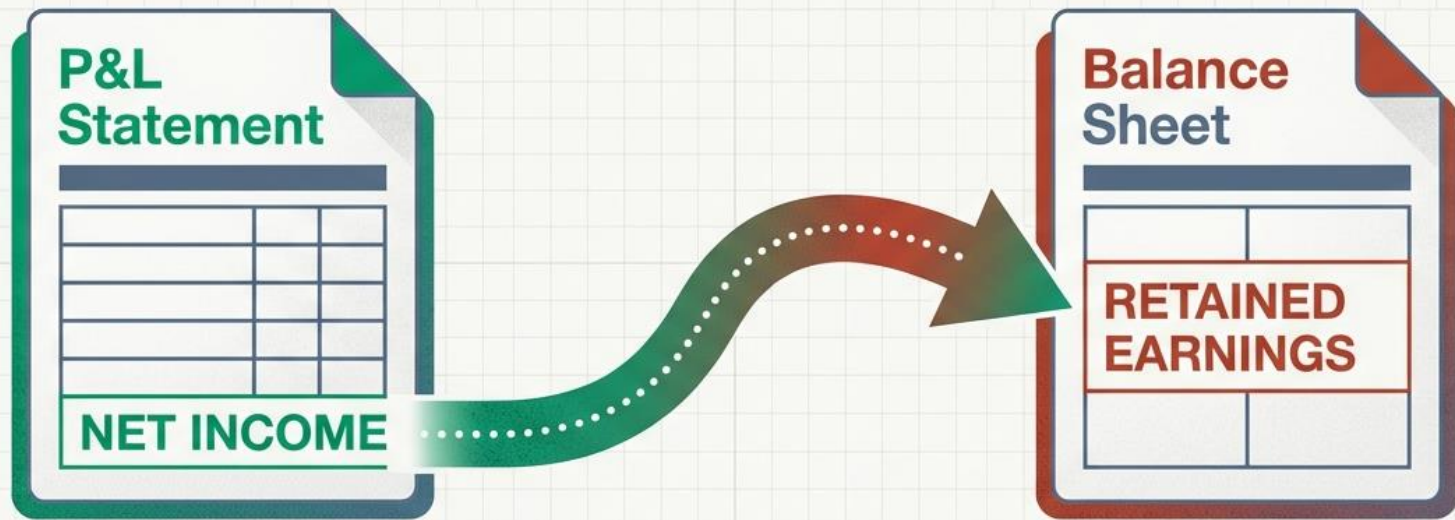


Trial Balance $\neq$ Balance Sheet

A Trial Balance is an internal, granular accounting check. The Balance Sheet is a formal, external report aggregated for investors (GAAP/IFRS).



Connecting the Dots: How the Movie becomes the Snapshot



These documents are linked. The P&L shows performance for *this year*. That Net Income flows into the Balance Sheet to become Retained Earnings, adding to the cumulative wealth of *all previous years*.

Which tool should you use?

Use the **P&L Statement** when...

- ✓ Analyzing profit margins
- ✓ Identifying trends in revenue or spending
- ✓ Cutting operational costs
- ✓ Determining tax liability

Use the **Balance Sheet** when...

- ✓ Evaluating company net worth
- ✓ Checking liquidity for short-term bills
- ✓ Analyzing debt capacity (Leverage)
- ✓ Tracking inventory levels

References

- <https://www.investopedia.com/ask/answers/121514/what-difference-between-pl-statement-and-balance-sheet.asp>
- https://youtu.be/_dANzaInfvA?si=QzC0SfYsTZ9JOz4M

FINANCE AND ACCOUNTS

- **BUDGET**
- **DEFICIT BUDGET AND WHAT IT MEANS**
- **CAPITAL AND REVENUE EXPENDITURE**
- **DEPRECIATION AND INCOME TAX**

FINANCE AND ACCOUNTS

- **SURPLUS BUDGET**

1. PROFIT AFTER TAX
2. CARRY OVER OR DISBURSAL
3. EBIDTA

INVESTMENT DECISIONS

TECHNICAL EVALUATION.

FEASIBILITY

ABILITY

COMPETITION

NOVELTY

DIFFERENTIATION OR LOW COST

GRASS ROOT OR EXISTING INFRASTRUCTURE

UNVESTMENT DECISIONS

LOCATION

LAND ACQUISITION

OTHER HISTORICAL AND GEOGRAPHICAL FACTORS

AVAILABILITY OF MANPOWER

ENVIRONMENTAL CLEARANCE.

OTHER RULES AND REGULATIONS.

**CONSISTENCY WITH COMPANY PHILOSOPHY AND
CULTURE**

INVESTMENT DECISIONS

RETURN ON INVESTMENT

AVAILABILITY OF CASH FLOW.

BANKS NBFCs OR OTHER INSTITUTIONS.

BACKWARD AREA BENEFITS

TIME FRAME OF MONEY REQUIREMENT.

OPPORTUNITY COST.

PRODUCT LIFE CYCLE

Estimation.

Competetion.

External factors

Tie up with prospective customers

RISK ANALYSIS

Evaluate And Quantify.

Do or Die category.

Environmental Impact

Sustainability.

Intangibles- Patents, Value addition, Market Reputation.

Opportunity cost.

ACCOUNTING SYSTEMS

Tracking Expenditure

Single Entry

Double entry

Manual (Redundant)

Computerised.

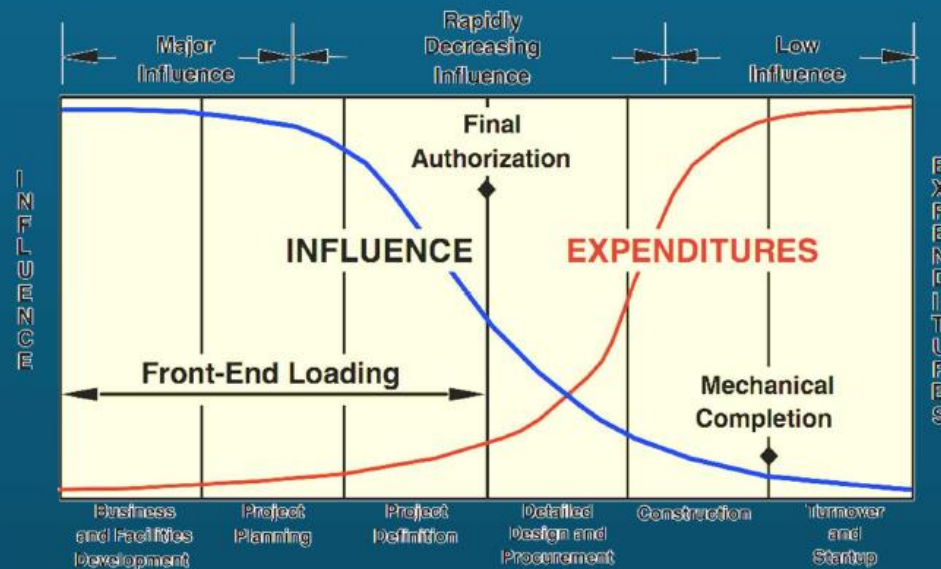
Companywide Software.

STEP COSTING DIAGRAM



Project Planning

Shape up of Project - FEL



CONTROLLING EXPENDITURE

- **Monitoring Expenditure:**
 1. Bar chart of Expenditure and activity
 2. Review of costs if required
 3. Prepare statements for accounts & Loan disbursal
 4. Control over runs beyond contingencies

A SYSTEM IS A GROUP OF INTERACTING OR INTERRELATED ELEMENTS THAT ACT ACCORDING TO A SET OF RULES TO FORM A UNIFIED WHOLE.

**QUALITY IS NOT THE RESPONSIBILITY OF A PARTICULAR
SET OF PERSONNEL KNOWN AS QC OR QA
BUT
A RESULT OF THE ENTIRE ORGANISATION FOLLOWING THE
STIPULATED SYSTEM.**

QUALITY SYSTEMS

**ANY QUALITY SYSTEM
ISO 14000 OR SIMILAR REQUIRES.**

**DEFINATION OF INPUTS
PROCESS
OUTPUT
AND
FEEDBACK**

- **Control of inputs:**

1. Process ,Log sheets, control tests. R.M Specification, operation manual
1. Equipment, Manpower skilled & semiskilled
2. Targets- yearly, Monthly, Daily

OPERATIONS

- **Control of process:**

Carry out process as per S.O.P

Annual plan-Monthly plan-weekly plan-Daily Plan-Hourly Plan-

What needs to be done now?

Define immediate action

Device own monitoring systems and plan personal schedule

- **Verify :**

Daily reports

Log sheet

Batch sheets

OPERATIONS

- **Monitor Output:**
- Remember Monitor Not Control
- Output Quality is a result of me meticulous follow up of S.O.P
- Prepare Statistical records
- Use of fuzzy logic

OPERATIONS

- **Safety :**
- People – Beware of daredevils & alarmists
- Environment
- Equipment

OPERATIONS

- **Reports:**

1. Daily, weekly, monthly, annual reports are written at Appropriate levels of hierarchy
2. Should contain financial numbers
3. Should contain safety record
4. Should contain reviews

OPERATIONS

- **Interactions:**

1. Purchase :

To decide stock levels, Decide Availability & shortage and fluctuation in prices

2. Maintenance :

To decide availability of manpower & equipment

OPERATIONS

- **Control of Effluents :**

1. Segregation
2. Chemical treatment
3. Biological treatment
4. Post treatment

OPERATIONS

- **Control of Costs:**

1. Raw Material
2. Utilities
3. Maintenance
4. Comparison and preparing and revising norms

OPERATIONS

- **Statutory Requirements:**
 1. Excise
 2. Sales tax
 3. Director industrial health & Safety
 4. Effluent control board
 5. Labor Departments

OPERATIONS

- **Working capital Requirement:**

- Define & Control

- Finished Goods Stocks

- Raw material stocks

- Goods in process & thereby define working capital

OPERATIONS

- **Improvement in process:**

- Adequate data

- Statistical analysis

- Interaction with design ,R&D

- Define changes & authorization

OPERATIONS

- **ISO Applicability:**
 1. Inputs
 2. Process
 3. Outputs
 4. Importance can not be understated

MARKETING

- **Marketing:**
 - Some misconception
 - Marketing and Sales
 - Customer base

MARKETING

- **Plans for Marketing:**

1. Immediate (Upto 3 Months)
2. Short term (1 year)
3. Long term (3-5 years)
4. Plan should include tonnages as well as revenges & budgets

MARKETING

- **Information:**

Products & Services

1. Uses
2. Specifications
3. Availability
4. Packaging
5. Regulatory issues

MARKETING

- **Pricing:**

Based on

Opportunity

Cost Based

Distress sale

MARKETING

- **Customers & Competitor**

1. Know your customers

- Policies

- Difficulties

- Requirements

2. Competitor

- Products

- Pricing

- Strength & weakness

- Practices.

MARKETING

- **Interactions:**

R & D for new products

Operations for plans/stocks

Sales for logistics

MARKETING

- **Information Gathering:**

- Library best source

- Trade fares

- Internal meetings

- Academic institutions

- Develop own networking

MARKETING

- **Remember**

You are

Face of the company

Control your Budget & overheads

Do not show off

Do not underwrite other Depts./personal

MARKETING

- **Inputs :**

- Production plan

- Product information / produces

- Pricing Guidelines

- Customer details

- **Outputs:**

- Marketing plan

- Review addition of customers

- Increase in sales , revenue

CAREER CHOICE

- It is normal to be uncertain
- Do not follow the crowd
- Is the interest in lighter studies genuine
- It is never too late to change
- Higher studies ? In which held, Job? In which field

WRITING AN SOP

- Be Honest & Realistic
- State what you feel not What you should feel
- A little uncertainty is not bad
- State what you intend to achieve through higher education
- Use understand language and informal narrative

- **Appearing for an interview :**

1. Get information about company & its product
2. For R& D be ready with some thoughts for the organization
3. Define what you think you will be required to do & contribute
4. Do not ask too many queries about locations and transfers
5. Do not be worried about slightly lower salary
6. Ask questions

- **Some Soft skills:**

1. Motivation- Maslow's theory
2. Communication- Johari window
3. Leadership –styles
4. Getting along with people –How ? , Why?
5. Recognizing Oneself

TOXICITY

- Toxicity of chemicals
- Concept of selective toxicity
- Types of Toxicity
 1. Oral
 2. Inhalation
 3. Dermal
 4. Neurotoxicity
 5. Carcinogens
 6. Teratogenicity

TOXICITY

TOXICITY :

- Acute & Subacute toxicity
- Impact of the above in working life

TOXICITY

- **Toxicity Of Chemicals:**

- Relation between
 - Toxicity & physical properties
 - Toxicity & Corrosivity
 - General Antidotes
 - Flammability flash point & its impact on chemical process
- Residues and Persistence –Why both are important
- Metabolism in human body'
- Aquatic toxicity and toxicity to honey bees
- Long term precautions in chemical industry
 - Do not be afraid & Do not be a daredevil
 - Trust your instinct
 - Take simple precautions

TOXICITY

Methods of Disposal of toxic chemicals:

1. Transportation
2. Storage
3. Handling

!! THANK YOU !!